

Assignment 03

Group Members:

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Submitted to : Dr.Saima

Outcomes Assessed:

CLO 2: Develop conceptual, logical, and physical database schemas by applying different data models to effectively design database systems

CLO 3: Analyse functional dependencies and resolve database anomalies through normalization techniques, ensuring efficient and reliable data organization.

Criteria	Completeness	Originality	Individual/ Team Effort	Presentation Skills/ Writing Skills	Timeliness	Final Score
Total Marks	8	2	4	4	2	20
Q#1 (CLO-2)						
Q#2 (CLO-3)						

Instructions;

Add your submitted EER/ERD, marked ERD and updated ERD in this assignment. Use the colored images of marked ERD in soft form.

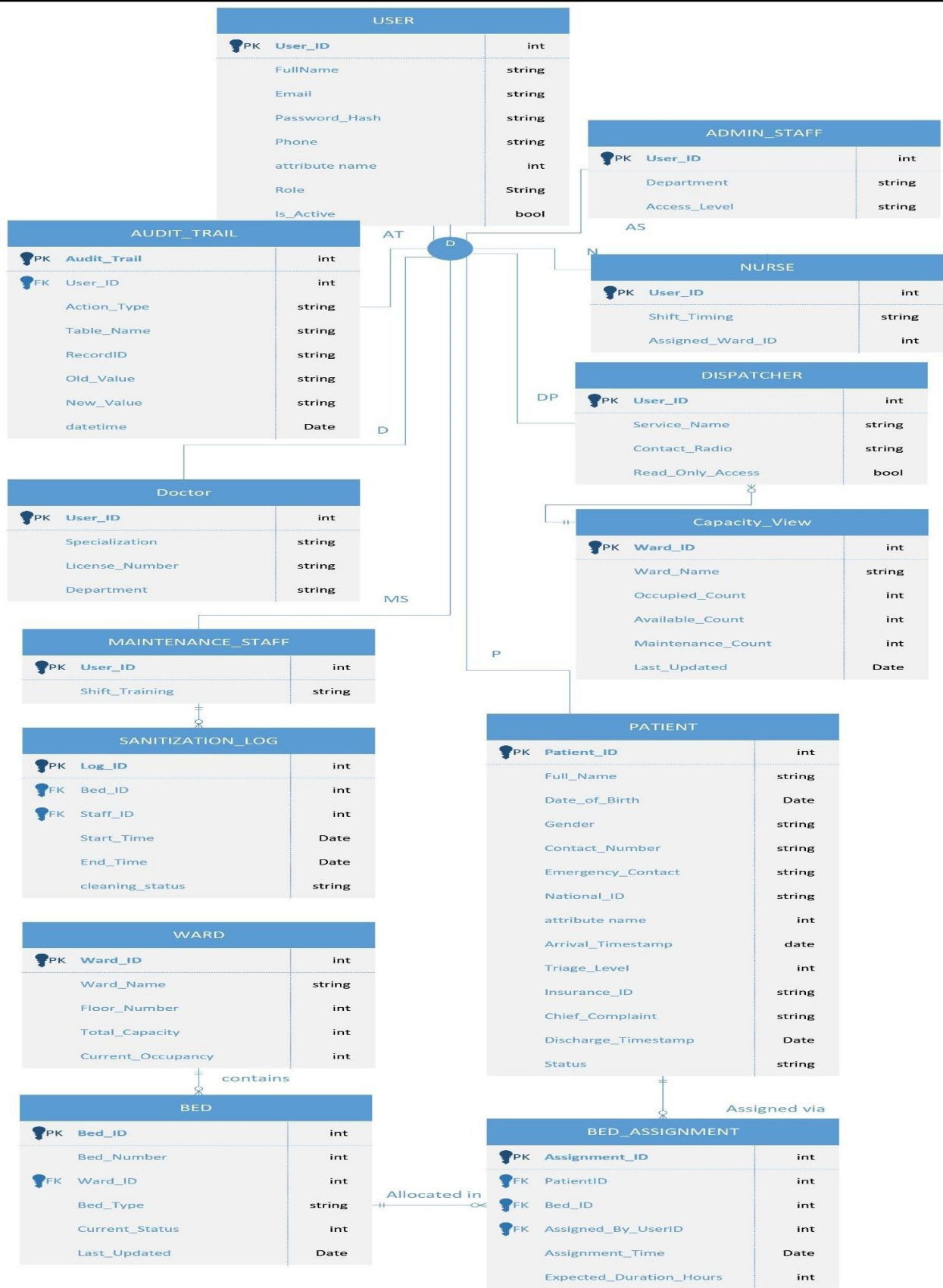
Conceptual and Logical Modeling (WP3)

Provide an EER/ERD diagram using Microsoft Visio to represent your database schema.

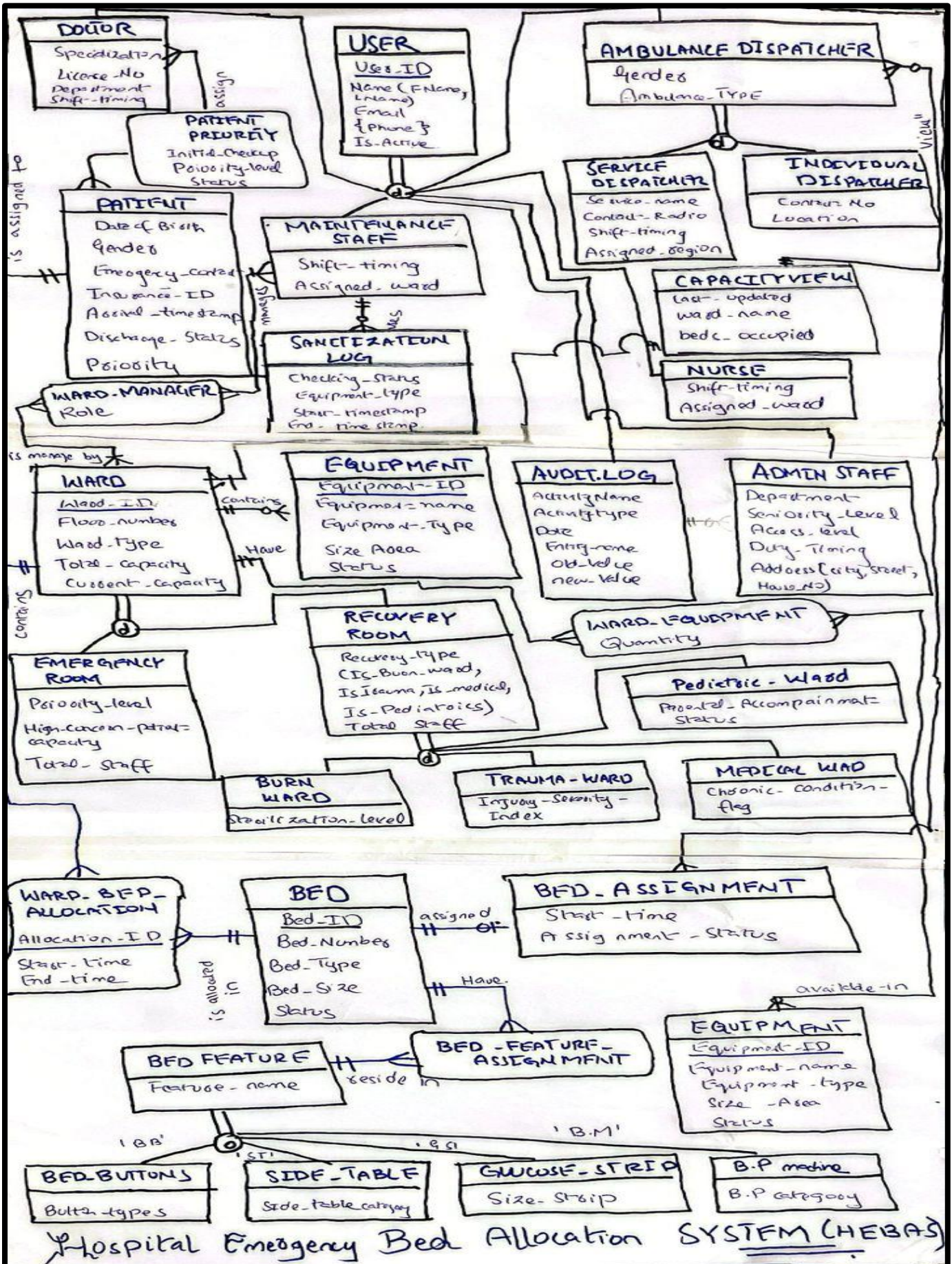
[EER/ERD diagram to be attached as produced in Microsoft Visio]

EER Diagram of HEBAS:

OLD EERD



UPDATED EERD



Hospital Emergency Bed Allocation SYSTEM (HEBAS)

How the Updated EERD Resolves These Loopholes

Your second diagram does a great job of turning this into a highly normalized, enterprise-grade schema. Here is how it fixes every single issue:

1. Separation of Concerns via Advanced Disjoint Specialization

Instead of forcing every staff member through a messy web of lines to the USER table, you implemented clean, explicit subclassing. USER now cleanly specializes into DOCTOR, MAINTENANCE STAFF, AMBULANCE DISPATCHER (which further breaks down into Service vs. Individual), NURSE, and ADMIN STAFF. This eliminates the messy routing lines and ensures permissions/roles are perfectly isolated.

2. Elimination of Redundant Paths with WARD_BED_ALLOCATION

You removed the confusing direct lines between WARD and BED. In their place, you introduced a clear associative entity/bridge: WARD_BED_ALLOCATION (with Allocation_ID, Start-Time, and End-Time). This cleanly normalizes the relationship, allowing you to track exactly when and how long beds are allocated to specific wards over time without hardcoding a permanent Ward_ID into the BED table.

3. Deep Specialization of Emergency Ward Types

To solve the generic "Ward" loophole, the updated EERD introduces a comprehensive disjoint subclass strategy for specialized rooms. WARD connects directly to specific room types like EMERGENCY ROOM and RECOVERY ROOM. The RECOVERY ROOM then branches into highly specific emergency categories:

- **BURN WARD** (Tracks unique attributes like Sterilization-level)
- **TRAUMA WARD** (Tracks Injury-Severity-Index)
- **MEDICAL WARD** (Tracks Chronic-Condition-flag)
- **PEDIATRIC WARD** (Tracks Parental-Accompaniment-Status)

4. Introduction of a Dedicated EQUIPMENT & Asset Management Subsystem

The updated diagram completely resolves the vague, handwritten asset notes from the first version by adding a robust equipment tracking framework. You created an EQUIPMENT entity (Equipment_ID, Name, Type, Status) linked to wards via WARD_EQUIPMENT (capturing Quantity). This ensures the system knows exactly what machinery is available where.

5. Granular Bed Customization (BED FEATURE & Subclasses)

Beds in an ER aren't identical. You solved this by mapping BED to a BED_FEATURE_ASSIGNMENT bridge table that hooks into a BED FEATURE superclass. This cleanly categorizes specific medical attachments using category codes (BB, ST, GS, BM) for:

- **BED BUTTONS** (Button-types)
- **SIDE TABLE** (Side-table-category)
- **GLUCOSE STRIP** (Size-Strip)
- **B.P. MACHINE** (B.P. Category)

6. Optimized Triage and Patient Flow (PATIENT PRIORITY)

Instead of a flat, chaotic link between patients and assignments, you introduced a PATIENT PRIORITY bridge entity directly between DOCTOR, PATIENT, and the system. This allows the emergency system to log an Initial-Checkup, assign a Priority-level-status, and route the patient dynamically based on urgency before they are ever assigned a physical bed.

7. Clean Log Isolation (AUDIT LOG & SANETIZATION LOG)

The chaotic audit pathways from diagram one are entirely cleaned up:

- AUDIT LOG is now a clean, standalone tracking entity linked directly to system actions, capturing ActivityName, ActivityType, Old-value, and New-value.
- SANETIZATION LOG (labeled as *Sanetization* in diagram 2) is cleanly mapped directly to MAINTENANCE STAFF, tracking timestamps and Equipment_type accurately without muddying up the primary patient care pathways.

Question 01 :Convert the EE/ERD of your project into relational data model.

Answer:

RELATIONAL DATA MODEL:

User:

<u>UserID</u>	FName	LName	Email	Is_Active
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User Contact:

<u>UserID</u>	<u>Phone</u>
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Doctor:

<u>DUserID</u>	Specialization	License_No	Department	Shift_Timing
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MAINTENANCE_STAFF

<u>MSUserID</u>	Shift_Timing	Assigned_Ward
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ADMIN_STAFF:

<u>ASUserID</u>	Department	Seniority_Level	Access_Level	Duty_Timing	City	Street	House_No
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AMBULANCE_DISPATCHER:

<u>ADUserID</u>	Gender	Ambulance_Type
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PATIENT;

<u>PUse rID</u>	Date_of_ Birth	Gen der	Emergency_ Contact	Insuran ce_ID	Arrival_Tim estamp	Discharge_ Status	Prio rity	Doctor_U serID	Ward_Man ager_ID
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AUDIT_LOG:

<u>Log_ID</u>	ActivityName	ActivityType	Date	Entry_Name	Old_Value	New_Value
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Nurse:

<u>NUserID</u>	Shift_Timing	Assigned_Ward
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AMBULANCE_DISPATCHER:

<u>UserID</u>	Gender	Ambulance_Type	<u>ViewID</u>
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SERVICE_DISPATCHER

<u>SUserID</u>	Service_Name	Contact_Radio	Shift_Timing	Assigned_Region
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INDIVIDUAL_DISPATCHER

<u>IUserID</u>	Contact_No	Location
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CAPACITY VIEW:

<u>ViewID</u>	Last_Updated	<u>Ward_Name</u>	Beds_Occupied
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WARD:

<u>Ward_ID</u>	Floor_Number	Ward_Type	Total_Capacity	Current_Capacity	<u>ViewID</u>
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WARD MANAGER:

<u>Ward_ID</u>	<u>UserID</u>	Role
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MAINTENANCE_STAFF

<u>UserID</u>	Shift_Timing	Assigned_Ward
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SANITIZATION_LOG:

<u>Sanitization_ID</u>	Checking_Status	Equipment_Type	Start_TimeStamp	End_TimeStamp	<u>UserID (Maintenance)</u>
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WARD:

<u>Ward_ID</u>	Floor_Number	Ward_Type	Total_Capacity	Current_Capacity
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EMERGENCY_ROOM

<u>EWard_ID</u>	Priority_Level	High_Concern_Patient_Capacity	Total_Staff
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RECOVERY_ROOM

<u>RWard_ID</u>	Recovery_Type	Is_Burn_Ward	Is_Trauma_Or_Medical	Is_Pediatrics	Total_Staff
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BURN_WARD

<u>BWard_ID</u>	Sterilization_Level
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TRAUMA_WARD

<u>TWard_ID</u>	Injury_Severity_Index
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MEDICAL_WARD

<u>MWard_ID</u>	Chronic_Condition_Flag
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PEDIATRIC_WARD

<u>PWard_ID</u>	Parental_Accompaniment_Status
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WARD_EQUIPMENT

<u>Ward_ID</u>	<u>Equipment_ID</u>	Quantity
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EQUIPMENT

<u>Equipment_ID</u>	Equipment_Name	Equipment_Type	Size_Area	Status
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WARD_BED_ALLOCATION:

<u>Ward_ID</u>	<u>Bed_ID</u>	<u>Allocation_ID</u>	Start_Time	End_Time
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BED:

<u>Bed_ID</u>	Bed_Number	Bed_Type	Bed_Size	Status	Ward_ID
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PATIENT;

User ID	Date_of_Birth	Gen der	Emergency_Contact	Insuran ce_ID	Arrival_Tim estamp	Discharge_ Status	Prio rity	Doctor_U serID	Ward_Mana ger_ID	Bed-Assign IP
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BED_ASSIGNMENT:

<u>Bed_Assignment-ID</u>	Start_Time	Assignment_Status	PatientID	<u>Bed_ID</u>
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BED:

<u>Bed_ID</u>	Bed_Number	Bed_Type	Bed_Size	Status	Ward_ID
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BED_FEATURE_ASSIGNMENT:

<u>Bed_ID</u>	<u>Feature_Name</u>
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BED_FEATURE

<u>Feature_Name</u>

BED_BUTTONS

<u>BFeature_Name</u>	Button_Types
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SIDE_TABLE

<u>STFeature_Name</u>	Side_Table_Category
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GLUCOSE_STRIP

<u>GSFeature_Name</u>	Size_Strip
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BP_MACHINE

<u>BMFeature_Name</u>	B.P_Category
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NORMALIZATION

User:

<u>UserID</u>	FName	LName	Email	Is_Active
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: 3NF

User Contact:

<u>UserID</u>	Phone
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: 3NF

Doctor:

<u>DUserID</u>	Specialization	License_No	Department	Shift_Timing
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: 3NF

MAINTENANCE_STAFF

<u>MSUserID</u>	Shift_Timing	Assigned_Ward
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: 3NF

ADMIN_STAFF:

<u>ASUserID</u>	Department	Seniority_Level	Access_Level	Duty_Timing	City	Street	House_No
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: 3NF

AMBULANCE_DISPATCHER:

<u>ADUserID</u>	Gender	Ambulance_Type
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: 3NF

PATIENT;

<u>PUse rID</u>	Date_of Birth	Gen der	Emergency_ Contact	Insuran ce_ID	Arrival_Tim estamp	Discharge_ Status	Prio rity	Doctor_U serID	Ward_Man ager_ID
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Solved Further

AUDIT_LOG:

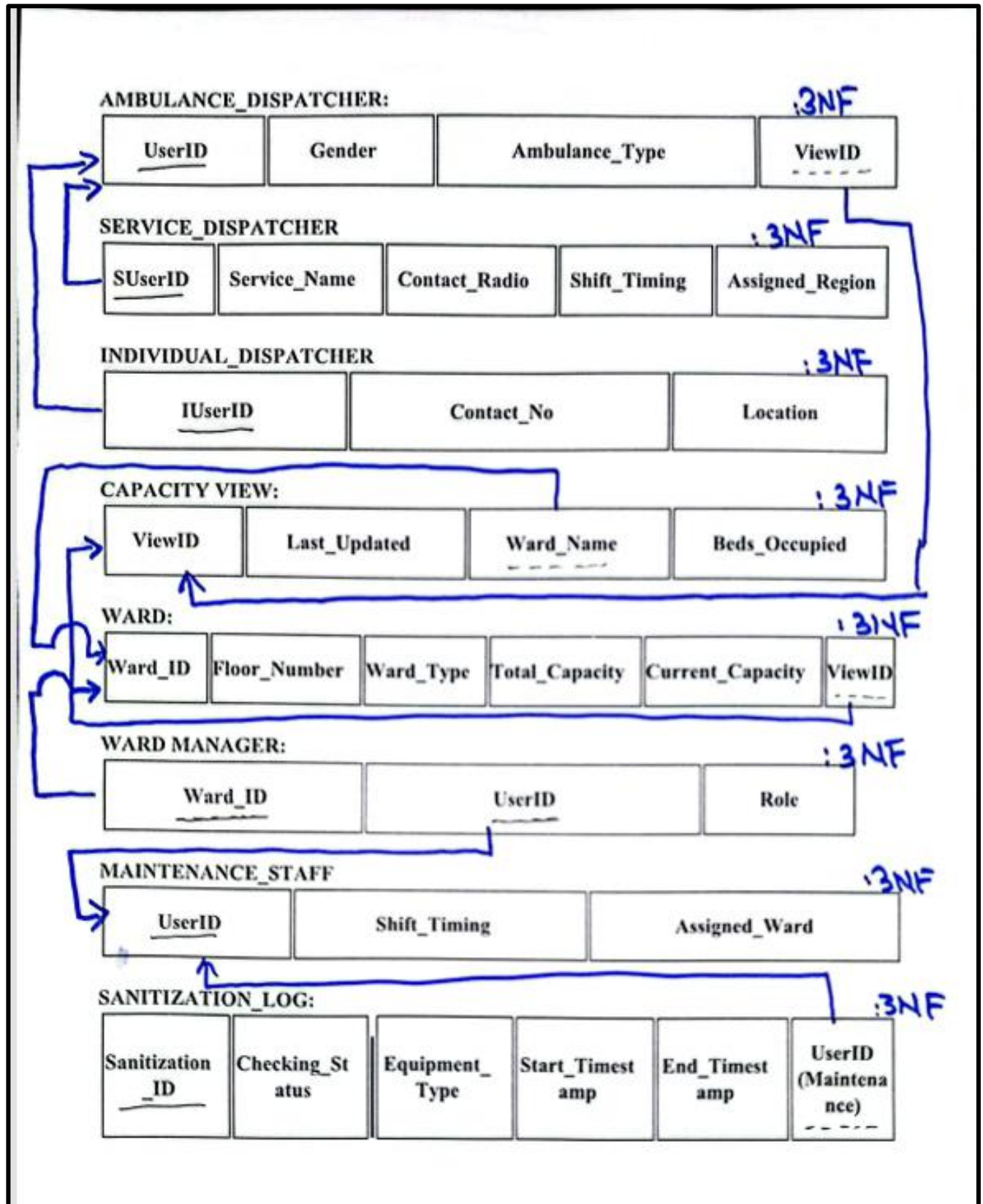
<u>Log_ID</u>	ActivityName	ActivityType	Date	Entry_Name	Old_Value	New_Value
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: 3NF

Nurse:

<u>NUserID</u>	Shift_Timing	Assigned_Ward
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: 3NF



WARD: ↓

<u>Ward_ID</u>	Floor_Number	Ward_Type	Total_Capacity	Current_Capacity
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:3NF

EMERGENCY_ROOM

<u>EWard_ID</u>	Priority_Level	High_Concern_Patient_Capacity	Total_Staff
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:3NF

RECOVERY_ROOM

<u>RWard_ID</u>	Recovery_Type	Is_Burn_Ward	Is_Trauma_Or_Medical	Is_Pediatrics	Total_Staff
-----------------	---------------	--------------	----------------------	---------------	-------------

:3NF

BURN_WARD

<u>BWard_ID</u>	Sterilization_Level
-----------------	---------------------

:3NF

TRAUMA_WARD

<u>TWard_ID</u>	Injury_Severity_Index
-----------------	-----------------------

:3NF

MEDICAL_WARD

<u>MWard_ID</u>	Chronic_Condition_Flag
-----------------	------------------------

:3NF

PEDIATRIC_WARD

<u>PWard_ID</u>	Parental_Accompaniment_Status
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:3NF

WARD_EQUIPMENT

<u>Ward_ID</u>	<u>Equipment_ID</u>	Quantity
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:3NF

EQUIPMENT ↓

<u>Equipment_ID</u>	Equipment_Name	Equipment_Type	Size_Area	Status
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:3NF

WARD_BED_ALLOCATION:

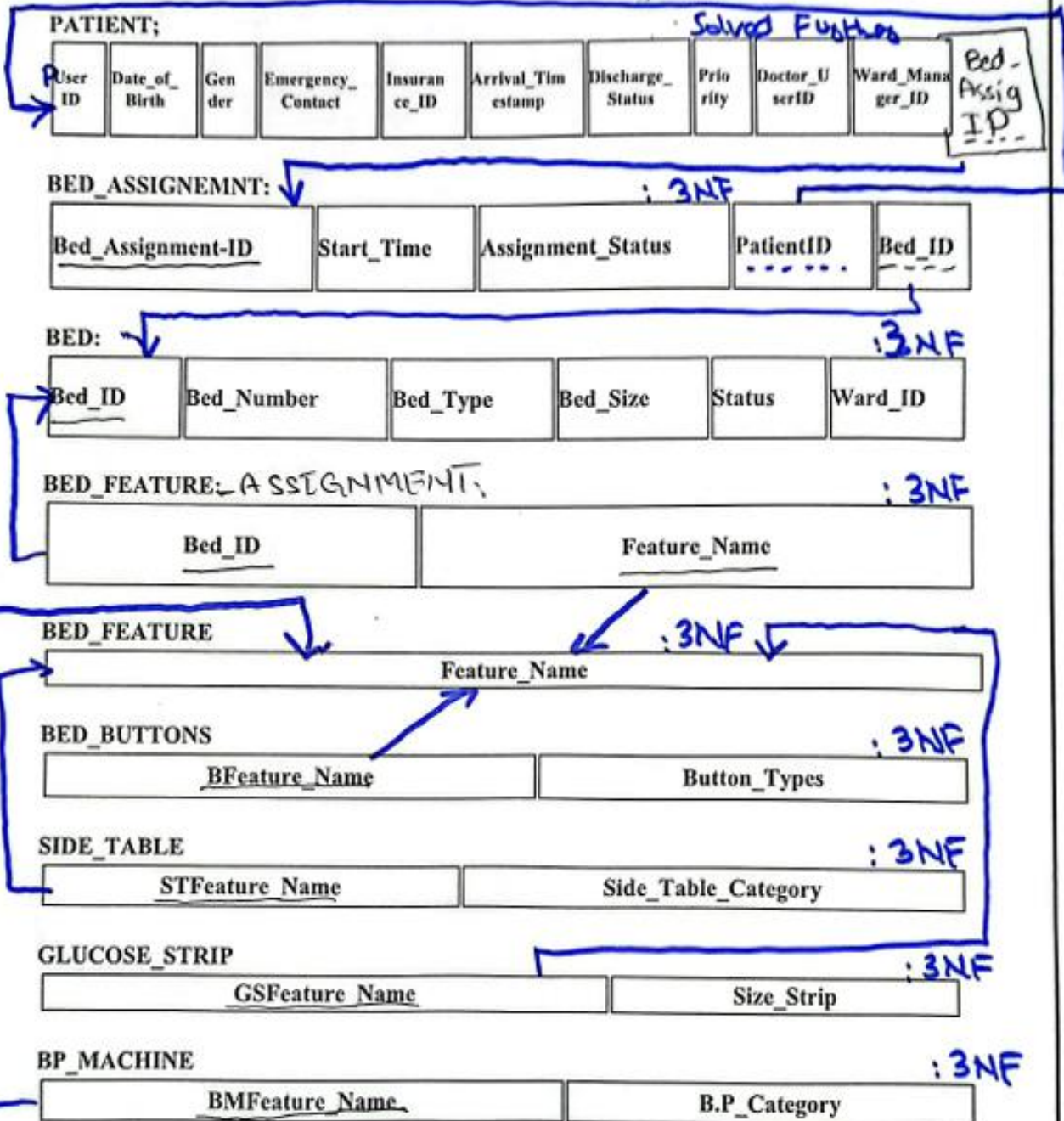
<u>Ward_ID</u>	<u>Bed_ID</u>	<u>Allocation_ID</u>	Start_Time	End_Time
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:3NF

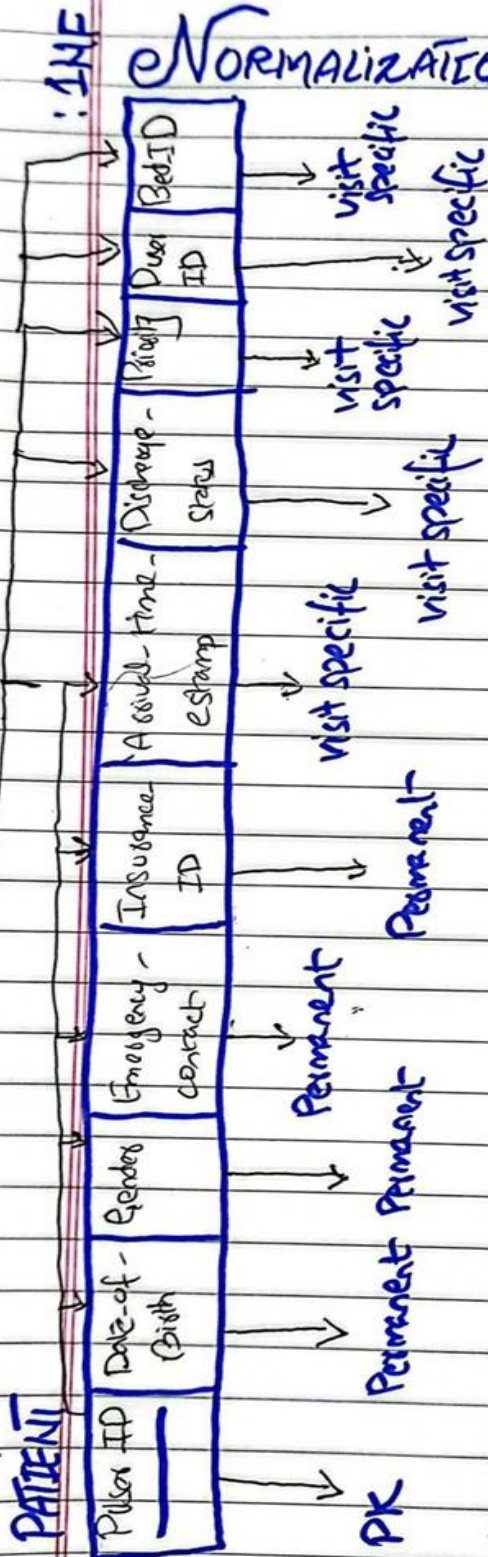
BED: ↓

<u>Bed_ID</u>	Bed_Number	Bed_Type	Bed_Size	Status	<u>Ward_ID</u>
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:3NF



Date: _____



NORMALIZATION:

What is the problem? The problem is the actual Patient-profile data remain constant but the (Admission-time-stamp, Discharge-status, Priority, Doctor-ID, Bed-ID) varies every time when the patient visits so to make it more better we have to decompose it further, like Patient-profile or Admission.

: 3NF

PATIENT - PROFILE:

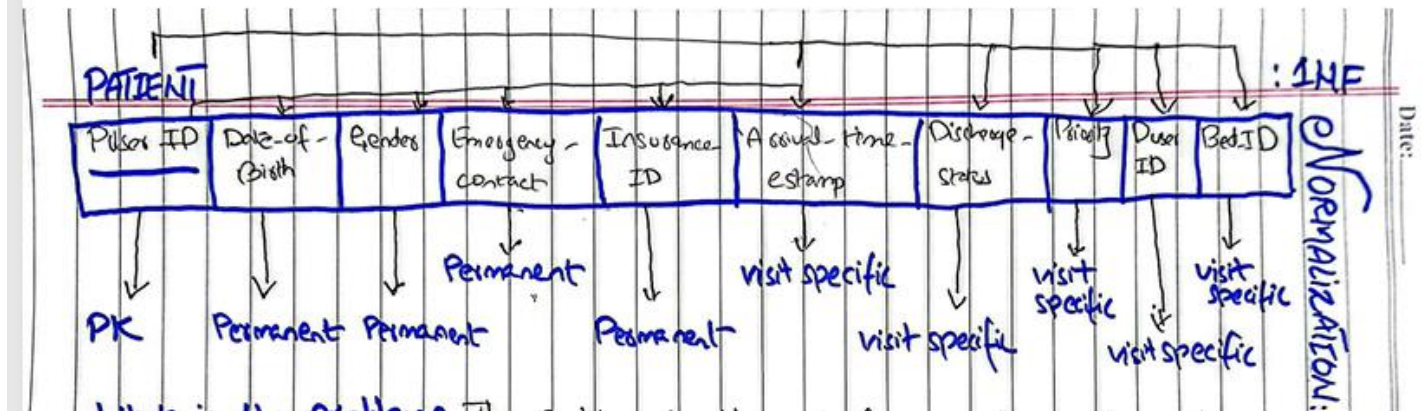
Patient-ID	Date-of-Birth	Gender	Emergency-Contact	Insurance-ID
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: 3NF

ADMISSION:

Admission-ID	Patient-ID	Admission-time-stamp	Discharge-status	Priority	Doctor-ID	Bed-ID
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Original Relation:

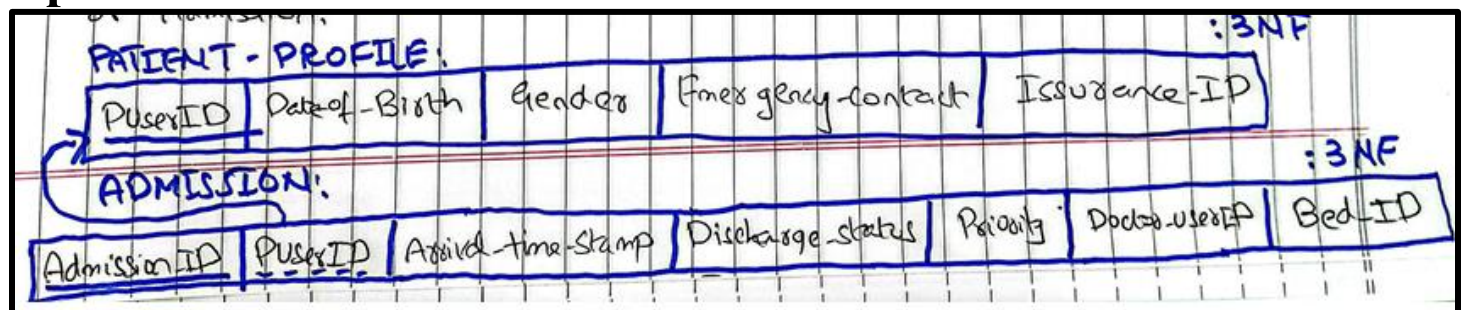


Reason of Normalization:

What is the problem? The problem is the actual patient-profile data remain constant but the (Admission-time-stamp, Discharge-status, Priority, Doctor-ID, Bed-ID) varies every time when the patient visits so to make it more better we have to decompose it further. Like Patient Profile or Admission.

: 3NF

Updated Normalization:



Explanation of Normalization:

Relational Schema 3NF Justifications

Definition of Third Normal Form (3NF)

A relation is in Third Normal Form (3NF) if:

1. It is already in Second Normal Form (2NF), meaning every non-key attribute is fully functionally dependent on the entire primary key.
2. It contains no transitive dependencies, meaning no non-key attribute depends on another non-key attribute.

Part 1: 3NF Justifications for Each Relation

1. USER

Primary Key: UserID

The USER relation is in 3NF because all descriptive attributes, including first name, last name, email, password hash, and account status, depend directly and solely on UserID. No non-key attribute determines another non-key attribute, so there are no transitive dependencies.

2. USER_CONTACT

Primary Key: (UserID, Phone_Number) or ContactID (depending on implementation)

This relation separates multi-valued phone numbers from the USER entity to preserve normalization. Each phone number belongs to a specific user and depends entirely on the identifying key. No non-key attribute determines another non-key attribute.

3. DOCTOR

Primary Key: DUserID

The DOCTOR relation stores doctor-specific information linked to a generalized user profile. Attributes such as specialization, medical license number, department, and shift schedule depend entirely on DUserID. None of these attributes functionally determine one another, eliminating transitive dependencies.

4. MAINTENANCE_STAFF

Primary Key: MSUserID

All attributes, including assigned ward area and operational shift timings, depend directly on MSUserID. No non-key attribute determines another non-key attribute, ensuring compliance with 3NF.

5. ADMIN_STAFF

Primary Key: ASUserID

Administrative attributes such as department, seniority level, security clearance, and address information are fully dependent on ASUserID. Since these attributes are independent of one another, the relation satisfies 3NF.

6. AMBULANCE_DISPATCHER

Primary Key: ADUserID

Dispatcher-specific information, including assigned ambulance category and operational details, depends entirely on ADUserID. No transitive dependency exists among non-key attributes.

7. SERVICE_DISPATCHER

Primary Key: SUserID

Attributes such as radio frequency, regional assignment, and shift timings depend solely on SUserID. These attributes are independent and do not determine one another.

8. INDIVIDUAL_DISPATCHER

Primary Key: IUserID

Communication details and location assignments are fully dependent on IUserID. No transitive dependency exists among non-key attributes.

9. AUDIT_LOG

Primary Key: Log_ID

The AUDIT_LOG relation records system events using a surrogate primary key. Attributes such as timestamps, action types, target attributes, old values, and new values depend entirely on Log_ID. Historical values do not determine each other, ensuring 3NF compliance.

10. NURSE

Primary Key: NUserID

Shift schedules and ward assignments depend directly on NUserID. Since shift timing does not determine ward assignment, there are no transitive dependencies.

11. CAPACITY_VIEW

Primary Key: ViewID

Capacity metrics such as ward name, available beds, and synchronization timestamps depend solely on ViewID. No non-key attribute determines another non-key attribute.

12. WARD

Primary Key: Ward_ID

Ward-specific attributes such as floor number, ward type, and capacity depend directly on Ward_ID. These attributes are independent of one another, satisfying 3NF.

13. WARD_MANAGER

Primary Key: UserID

Managerial role information and associated ward assignments depend entirely on UserID. No non-key attribute determines another non-key attribute.

14. SANITIZATION_LOG

Primary Key: Sanitization_ID

Cleaning status, equipment category, timestamps, and technician assignments all depend solely on Sanitization_ID. There are no transitive dependencies.

15. EMERGENCY_ROOM**Primary Key:** EWard_ID

Emergency room attributes such as triage level, surge capacity, and emergency staff count depend directly on EWard_ID. None of these attributes determines another.

16. RECOVERY_ROOM**Primary Key:** RWard_ID

Recovery room properties such as pediatric support flags and staff counts depend entirely on RWard_ID. These attributes are independent and therefore satisfy 3NF.

17. BURN_WARD**Primary Key:** BWard_ID

The specialized attribute Sterilization_Level depends entirely on BWard_ID. Since no additional non-key attributes exist, the relation naturally satisfies 3NF.

18. TRAUMA_WARD**Primary Key:** TWard_ID

The Injury_Severity_Index attribute depends directly on TWard_ID. With no additional non-key dependencies present, the relation is in 3NF.

19. MEDICAL_WARD**Primary Key:** MWard_ID

The Chronic_Condition_Flag attribute depends entirely on MWard_ID. No transitive dependencies exist.

20. PEDIATRIC_WARD**Primary Key:** PWard_ID

The Parental_Accompaniment_Status attribute is fully dependent on PWard_ID. Since there are no additional non-key dependencies, the relation satisfies 3NF.

21. WARD_EQUIPMENT**Primary Key:** (Ward_ID, Equipment_ID)

This associative entity stores the quantity of specific equipment within a ward. The Quantity attribute depends on the complete composite key and not on either component individually. Therefore, the relation satisfies 3NF.

22. EQUIPMENT

Primary Key: Equipment_ID

Attributes such as equipment name, category, dimensions, and operational status depend directly on Equipment_ID. No transitive dependency exists.

23. WARD_BED_ALLOCATION**Primary Key:** Allocation_ID

Allocation timestamps, ward references, and bed references depend fully on Allocation_ID. Check-in and check-out times are properties of the allocation event itself and do not determine other attributes.

24. BED**Primary Key:** Bed_ID

Attributes including bed label, model type, dimensions, and status depend entirely on Bed_ID. None of these attributes determines another.

25. BED_ASSIGNMENT**Primary Key:** Bed_Assignment_ID

Assignment details such as patient reference, bed reference, assignment status, and timestamps depend fully on Bed_Assignment_ID. No transitive dependencies exist.

26. BED_FEATURE_ASSIGNMENT**Primary Key:** (Bed_ID, Feature_ID)

This bridge relation maps features to beds. Since it contains only linking attributes and no additional dependent attributes, it naturally satisfies 3NF.

27. BED_FEATURE**Primary Key:** Feature_ID

The attribute Feature_Name depends solely on Feature_ID. No additional dependencies exist.

28. BED_BUTTONS**Primary Key:** Feature_ID

The Button_Types attribute depends entirely on the associated feature identifier. No transitive dependency exists.

29. SIDE_TABLE**Primary Key:** Feature_ID or Side_Table_ID

The attribute Side_Table_Category depends fully on the entity's primary key. No additional dependencies are present.

30. GLUCOSE_STRIP

Primary Key: Feature_ID or Strip_ID

The Size_Strip attribute depends entirely on the strip identifier. No non-key attribute determines another.

31. BP_MACHINE

Primary Key: Feature_ID or BP_Machine_ID

The BP_Category attribute depends directly on the primary key. Since no additional non-key attributes exist, the relation satisfies 3NF.

Part 2: Detailed Explanation of Patient Decomposition

Problem in the Original Design

Initially, all patient information was stored in a single table that combined:

- Static demographic data (Date_of_Birth, Gender, Emergency_Contact, Insurance_ID)
- Dynamic hospital visit data (Arrival_Timestamp, Bed_ID, Priority, Discharge_Status)

This structure created several normalization problems:

Redundancy

Whenever a patient visited the hospital multiple times, permanent demographic information had to be repeatedly stored for every admission.

Update Anomalies

Updating patient information, such as an emergency contact number, could leave inconsistent values across different visit records.

Insertion and Deletion Anomalies

A patient could not be added to the database without creating an admission record, and deleting the final admission record could unintentionally remove the patient's demographic data.

3NF-Based Decomposition

To eliminate redundancy and dependency anomalies, the original relation was decomposed into two separate entities:

PATIENT_PROFILE

Primary Key: PUserID

This relation stores static patient information that remains relatively constant over time:

- Date_of_Birth
- Gender
- Emergency_Contact

- Insurance_ID

Each patient's demographic data is stored only once.

ADMISSION

Primary Key: Admission_ID

Foreign Key: PUserID

This relation stores visit-specific operational information:

- Arrival_Timestamp
- Discharge_Status
- Priority
- Assigned_Doctor
- Bed_ID

Each hospital visit generates a separate admission record linked back to the patient profile through PUserID.

Benefits of the Decomposition

Reduced Redundancy

Static patient data is stored once instead of being repeated for every admission.

Improved Consistency

Updating patient details automatically reflects across all future admissions because the information exists in a single location.

Better Scalability

The database can efficiently support multiple admissions per patient without duplicating demographic information.

Full 3NF Compliance

- PATIENT_PROFILE contains only attributes dependent on PUserID.
- ADMISSION contains only attributes dependent on Admission_ID.
- No transitive dependencies exist between non-key attributes.

Therefore, the decomposition successfully transforms the original structure into a normalized 3NF design.
